JOHANSSON ENGINEERING

03 — What Can Go Wrong

For each vessel in your inventory (Doc 02), think about what hazards exist, what could cause them, and how bad it would be. Rate each risk and decide if you need to do something about it.

How to Rate a Risk

Probability (P) — how likely is it?

1	2	3	4	5
Very unlikely	Unlikely	Possible	Likely	Very likely

Consequence (C) — how bad would it be?

1	2	3	4	5
Negligible	Minor injury	Medical treatment	Serious injury	Fatal

Risk = P x C

1-4	5-9	10-15	16-25
Low — OK	Medium — monitor	High — fix it	Critical — stop

Vessel: Nr ____ ID: _____ Description: _____

#	WHAT CAN HAPPEN	WHY IT HAPPENS	HOW BAD IS IT	WHAT PROTECTS IT NOW	P	C	RISK
1							
2							
3							
4							
5							

Action needed? No / Yes: _____

Vessel: Nr ____ ID: _____ Description: _____

#	WHAT CAN HAPPEN	WHY IT HAPPENS	HOW BAD IS IT	WHAT PROTECTS IT NOW	P	C	RISK
1							
2							
3							
4							
5							

Action needed? No / Yes: _____

Vessel: Nr ___ ID: _____ Description: _____

#	WHAT CAN HAPPEN	WHY IT HAPPENS	HOW BAD IS IT	WHAT PROTECTS IT NOW	P	C	RISK
1							
2							
3							
4							
5							

Action needed? No / Yes: _____

(Print extra copies of this page for more vessels)

Common Hazards — Use This As a Starting Point

WHAT CAN HAPPEN	WHY IT HAPPENS	HOW BAD
Over-pressure / burst	Safety valve or control failure	Blast, equipment damage
Internal corrosion	Condensate, corrosive medium	Wall thins, eventual failure
External corrosion	Weather, dripping, insulation damage	Wall thins, eventual failure
Leakage	Failed gasket, cracked weld	Fluid release, scald risk
External impact	Vehicle, nearby works	Equipment damage
Safety valve stuck	Corrosion, seized mechanism	Over-pressure or release
High temperature	Control failure	Damage, fire risk

Risk Summary

NR	VESSEL ID	HIGHEST RISK	ACTION NEEDED
1		Low / Med / High	
2		Low / Med / High	
3		Low / Med / High	
4		Low / Med / High	
5		Low / Med / High	
6		Low / Med / High	
7		Low / Med / High	
8		Low / Med / High	
9		Low / Med / High	
10		Low / Med / High	

Sign-Off

Assessed by	
Signature	
Date	
Next review due	